

Business Requirement Document (BRD)

1. Project Overview

This project aims to develop an interactive Power BI dashboard for analyzing player performance in the ICC T20 World Cup. The dashboard provides deep insights into batting and bowling metrics, helping users evaluate and compare individual players and construct a custom "Final 11" team.

2. Business Objectives

- Enable performance-based player selection for various roles such as Openers, Anchors, Finishers, All-Rounders, and Bowlers.
 - Provide combined and individual player insights across batting and bowling.
 - Visualize metrics such as Strike Rate, Batting Average, Bowling Economy, Dot Ball %, and more.
 - Allow easy comparison between players from different teams based on key performance indicators.
 - Support decision-making for team combinations in fantasy leagues or analysis teams.
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3. Scope

In Scope:

- Player-level performance metrics
- Custom team selection interface
- Comparative visuals for batting and bowling stats
- Dashboard visuals across roles: Openers, Middle Order, Finishers, All-Rounders, Bowlers

Out of Scope:

- Real-time data feed

- Match predictions or simulations
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4. Data Sources

- **Tables:**
 - fact_batting_summary
 - fact_bowling_summary
 - dim_player
 - **Data Type:** Aggregated performance metrics per player across matches
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5. Key Measures & Calculations

Measures (Using DAX):

- Total Runs
- Total Innings Batted/Dismissed
- Batting Average
- Balls Faced
- Strike Rate
- Boundary %
- Batting Position
- Bowling Economy
- Bowling Average
- Bowling Strike Rate
- Wickets
- Dot Ball %
- Player Selection Flag

Calculated Columns:

- Boundary Runs (Batting & Bowling)
- Custom Batting Order (Final 11 logic using SWITCH)

Refer to DAX formulas section for exact syntax (included in a separate technical appendix).

6. Dashboard Features

Player Role Tabs:

- **Power Hitters / Openers**
- **Anchors / Middle Order**
- **Finishers / Lower Order**
- **All-Rounders**
- **Specialist Bowlers**

Visuals:

- Scatter plots for Strike Rate vs Batting Average
 - Bar/Column charts for performance comparison
 - Tooltip-enhanced tables for detailed metrics
 - Search and filter panel to select specific players
 - "Final 11" team composition view
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7. KPIs Tracked

- **Batting KPIs:** Strike Rate, Batting Average, Boundary %, Average Balls Faced
 - **Bowling KPIs:** Wickets, Economy, Bowling Strike Rate, Dot Ball %, Bowling Average
 - **Team KPIs:** Combined player metrics for selected roles
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8. User Roles

- **Analyst:** Uses dashboard for in-depth player comparisons
 - **Fantasy League Manager:** Builds teams based on actual performance data
 - **Cricket Enthusiast:** Explores trends and top performers
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9. Assumptions & Constraints

- Data is cleaned and pre-aggregated before being loaded into Power BI.
 - Players with no data are not considered in visuals.
 - Player role assignment is static and based on cricketing context.
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10. Appendices

- **Appendix A:** DAX Measure Definitions
- **Appendix B:** Dashboard Screenshots (See attached PDF: "Dashboard Images.pdf")